

Intelligent Dead Reckoning

Android application: status and roadmap
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Where the project stands. The Android application is complete and recording reliably. Dead reckoning with map-matched heading feedback works and is measured. Two further accuracy features are fully built but deliberately switched off, because both depend on one unresolved defect: while running without GNSS, the integrator recovers about 30% less distance than was actually travelled. That defect has now been diagnosed to a specific cause, and fixing it needs no machine-learning model and no external dependency.

1. What runs on the phone

A foreground service records eleven sensor streams and drives four estimators in parallel across three threads. There is exactly one writer thread, so no locking is needed anywhere in the recording path.

THREAD	OWNS	RATE
imu-logger	Sensor, GNSS and location callbacks; all file writes; the integrator	200 Hz
imu-ml	Model load and inference (about 40 ms per forward pass)	10 Hz
map-match	Mapsforge tile reads, HMM matcher, road graph	0.5 Hz

Every session writes **imu.csv**, **gps.csv**, **gnss_status.csv**, **deadreckon.csv**, **ml.csv**, **mapmatch.csv** and **session.json**. All timestamps share the monotonic elapsed-realtime clock, so they can be aligned exactly; session.json records the mapping to UTC. The map draws three tracks at once, GPS, IMU-only and map-snapped, so the divergence a tunnel would cause is visible live.

2. What works today

Application and interface (Task 4)

Essentially complete. Offline OpenStreetMap rendering, radius preloading, downloadable map zones for other parts of India, theme and settings screens, collapsible session card, and a locate button that zooms to a 1 km radius only when currently showing more. Records reliably on the Samsung S21 FE.

Dead reckoning with map-matched heading feedback

A strapdown inertial integrator free-runs whenever GNSS is unavailable. An online HMM map matcher snaps the drifting estimate to the road network and feeds the matched road bearing back as a heading observation. Measured across 46 to 57 synthetic outages per duration:

ESTIMATOR	10 s	30 s	60 s
Dead reckoning + heading feedback (CRSE)	10.2 m	36.0 m	87.5 m

On-device road topology

The strongest technical result so far. Mapsforge map files store ways clipped per tile with no node identity, so the network has no connectivity and the conventional answer is to route against an OSRM server, which an offline application cannot do. However, coordinates are stored in microdegrees, roughly 0.11 m, so segments that once shared an OpenStreetMap node still come back on identical coordinates. Snapping endpoints onto a

half-metre grid rebuilds the graph entirely on the device.

MEASURE	RESULT
Segments around IIT Kharagpur	25,383
Nodes recovered	18,570
Segments gaining at least one link	100.0%
Segments in a single connected component	98.5%
Mean node degree	2.73

No server, no additional download, and it works underground. Tile-boundary clipping heals for free, because both halves of a clipped way end on the same border coordinate.

Evaluation harness

CRSE, CAE and AEPS implemented to the same definitions the model side of the project uses, so results are directly comparable. Synthetic outages are cut from recorded sessions and scored against GPS ground truth.

3. Built, tested, and deliberately switched off

Two complete features are one line from being live. Both are disabled because measurement showed they would currently make the application worse, not better.

Along-road tracking

Walks position as a scalar along a road polyline, so the road supplies direction and heading error stops entering the answer altogether. This is the mechanism behind the accuracy of the server-based approach, reproduced without a server.

CONFIGURATION	10 s	30 s	60 s
Dead reckoning + heading feedback (baseline)	10.2 m	36.0 m	87.5 m
Along-road, fed the integrator's distance	12.6 m	40.8 m	104.7 m
Along-road, fed true distance	8.3 m	22.3 m	50.5 m

Given a trustworthy distance it cuts 60-second outage error by **42%**. Given the distance we can currently supply, it is 20% worse than doing nothing. The idea and the implementation are both sound; the input is not.

Two findings from building it are worth recording. Seeding matters: entering the road at the moment GNSS is lost, when position is still exact, is 42% better, whereas entering later from a drifted position is 20% worse. Same code, different starting point. And an earlier encouraging result was withdrawn after it failed to survive a denser sample; it had been measured on five outages.

Model speed fusion

The integrator can now accept the neural network's speed estimate, blended into velocity magnitude only and weighted by a scalar Kalman gain formed from the model's own reported variance. A confident stand-still prediction sets the target speed to zero. It is applied only while unaided, since GNSS velocity beats anything a phone-IMU network can offer.

The current checkpoint is not fit to steer anything. Scored against GPS speed over 3,584 on-device inferences from four recorded sessions:

MEASURE	MODEL	REFERENCE
Correlation with GPS speed	-0.224	higher is better

MEASURE	MODEL	REFERENCE
Root mean square error	2.566 m/s	1.277 m/s for a constant
Output range	0.02 to 0.69 m/s	truth spans 0.03 to 5.10 m/s
Reported speed while parked	0.681 m/s	true value 0.039 m/s

It is twice as bad as predicting a constant, anti-correlated with the truth, and cannot reach the upper half of the speed range. The cause is visible in the training data: **304 training windows for a 3,848,196-parameter network**, about 12,659 parameters per example. The targets themselves are fine, spanning 0 to 7.29 m/s with good spread. There is simply nowhere near enough recorded driving.

4. The blocking defect, and its cause

While free-running without GNSS, the integrator recovers about 30% less distance than was actually travelled. This is the single largest error source in the system, and it is what keeps both features above switched off.

OUTAGE DURATION	10 s	30 s	60 s
Distance shortfall	15.0%	21.3%	29.5%
Zero-velocity updates firing	1.9%	0.0%	1.4%
Samples dropped by the timing gate	0.0%	0.0%	0.0%

What was ruled out

- **The zero-velocity update is not the cause.** It was the leading hypothesis, and it is wrong. It fires on 0 to 2% of samples during outages, and at 30 seconds the shortfall is 21.3% while it fires on 0.0% of them. Disabling it entirely moves the 60-second shortfall only from 29.5% to 24.7%.
- **Dropped samples are not the cause.** The timing gate discards 0.0%.

What it actually is

A systematic deceleration. Over 46 sixty-second outages, the levelled acceleration projected onto the direction of travel averages **-0.0195 m/s²**, and is negative in **89%** of them. Over 60 seconds that alone removes 1.17 m/s, roughly a third of a 3.4 m/s cruise, which accounts for the entire shortfall. The shortfall also grows with duration exactly as a constant deceleration would.

Two supporting measurements point at where it comes from. The levelled vertical axis reads 9.7574 m/s² while the accelerometer magnitude averages 9.8864, so gravity is not aligned with the estimated vertical and leaks into the horizontal plane. And the assumed gravity constant, 9.8066 m/s², is 0.08 m/s² away from this device's own mean.

The structural fault: the horizontal bias states are learned only during stand-still, so they are estimated exactly when they are not needed and frozen exactly when they are. During a tunnel transit, nothing corrects them.

Whole sessions are unaffected, with a distance ratio of 0.992, because stand-still periods keep re-anchoring the estimate. The shortfall is specific to unanchored free-running, which is precisely the case the application exists for.

A separate and distinct failure was also found: in one near-stationary session, 39.8% of moving samples were wrongly called stationary and the integrator recovered 11 m of 93 m. That is a real defect, but it is not the cause of the general shortfall.

5. What to build next

Ordered by value per unit of effort. Item 1 is now well specified by the diagnosis above and depends on nobody else.

#	WORK	WHY IT MATTERS	DEPENDS ON
1	GNSS-aided horizontal bias observer	Learn the acceleration bias while GNSS is healthy, by comparing integrated velocity against GPS velocity, and carry it into the outage. Directly removes the -0.0195 m/s^2 term and therefore most of the 30% shortfall. Unlocks the 42% gain from along-road tracking.	Nothing
2	Record driving data at vehicle speed	Fixes the 304-window training set and gives the evaluation harness a sample that is not three walking sessions. Every measurement in this document would become more trustworthy.	Field time
3	Retrain and re-export the model	Then run the gate script; enable speed fusion only if it beats a constant, and confirm end-to-end position error improves.	Item 2
4	Enable the two switches	One line each, once items 1 and 3 pass their gates. Re-run the outage harness to confirm.	Items 1, 3
5	Road-test heading feedback live	It has only ever run in replay, never on the phone with real motion. A genuine gap before any demonstration.	Field time
6	Fix map download promotion	Completed downloads are not promoted into place, because the file is owned by the system download provider.	Nothing
7	Publish the offline map as a release asset	The 210 MB map file still needs uploading so the application can fetch it.	Manual upload

The verification commands already exist and both currently fail closed, which is the intended behaviour:

COMMAND	CHECKS
<code>python -m eval.dr_diagnostics <sessions></code>	Where the distance shortfall goes
<code>python -m eval.model_speed_eval <sessions></code>	Whether the model beats a constant
<code>python -m eval.outage_eval <sessions></code>	End-to-end outage position error

6. Notes for the team

- **The on-device road matcher is new capability worth sharing.** It solves offline, on the handset, what the model-side pipeline currently uses an OSRM server for. Road topology does not require a routing server.
- **The training splits are not reproducible.** The data preparation directory on the model side is not committed, so results cannot be independently checked.
- **IO-VNBD cannot validate this integrator.** It is CAN bus and GPS data from Coventry with no phone IMU, so it can benchmark the model but not the on-device dead reckoning. Metric definitions are shared; the datasets are not.
- **Reported numbers come from three walking-pace sessions.** Every outage in them falls below the 5.0 m/s start-speed gate used in the reference literature, so these results are comparable in definition but not directly in magnitude. This is the strongest argument for item 2 above.

All measurements in this document are reproducible from the committed evaluation harness against the recorded sessions. Figures were produced on 3 September 2026.